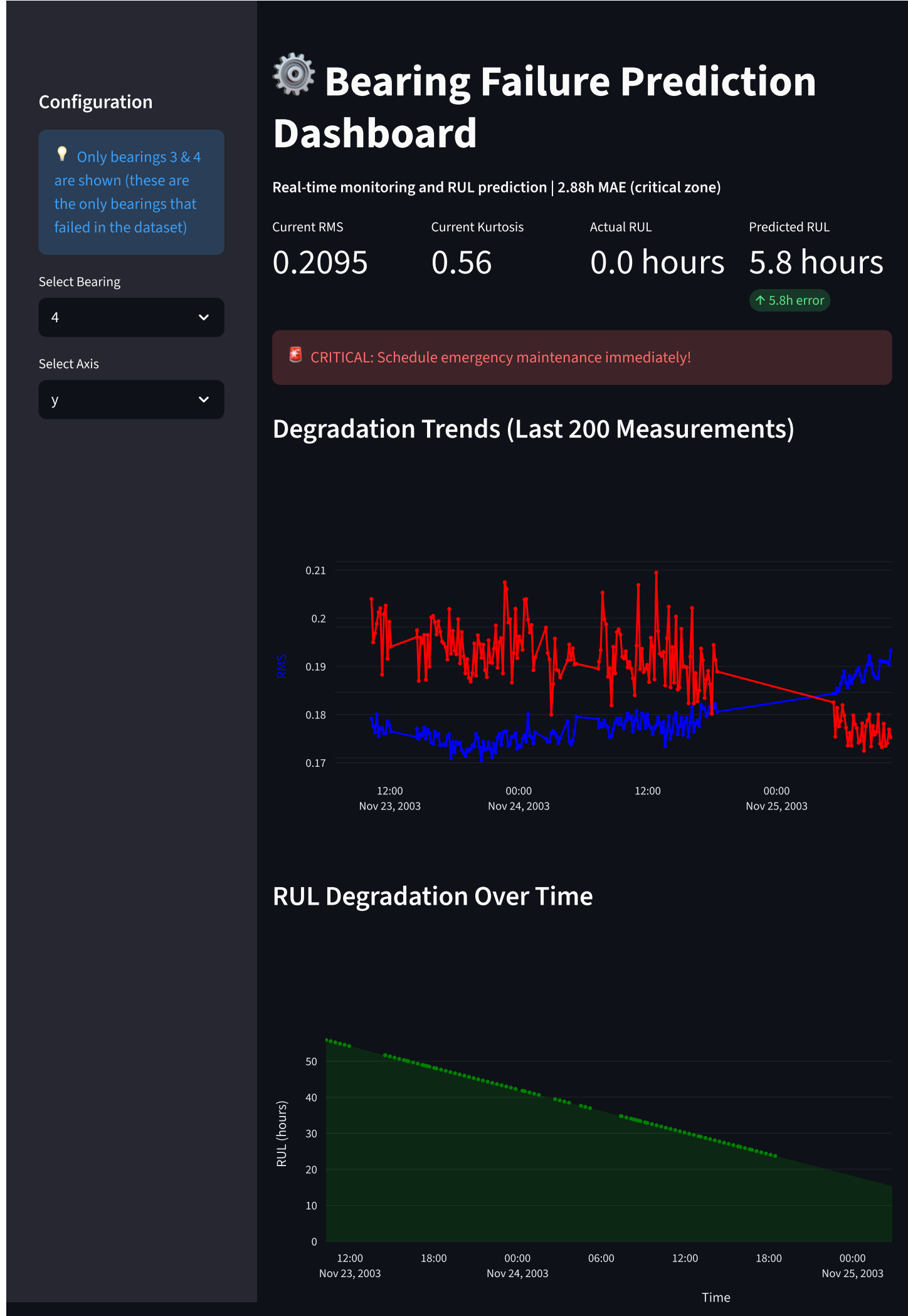
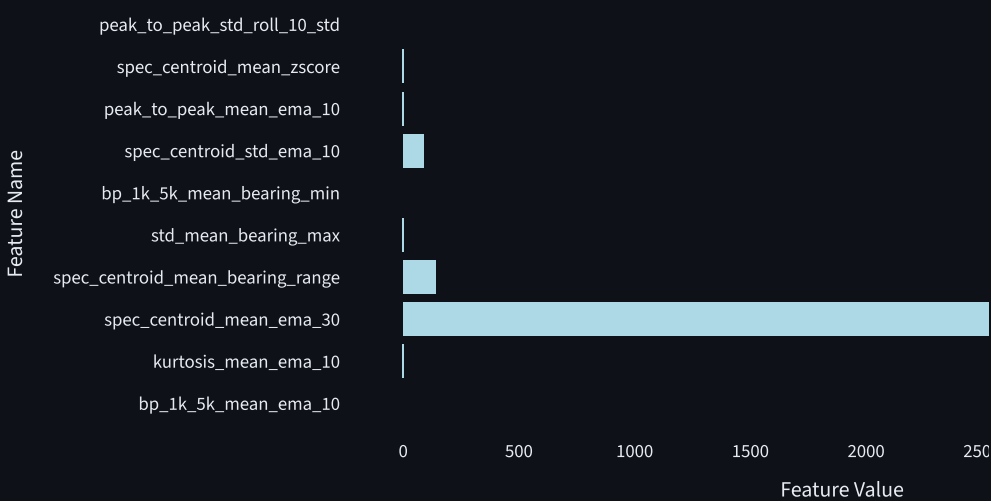




Top 10 Most Important Features (Current Values)

Feature Values at Current Time (Bearing 4-y)



Summary Statistics

Metric	Value	Metric	Value
Mean RMS	0.1790	Mean Kurtosis	1.20
Max RMS	0.2095	Max Kurtosis	1.92
Min RMS	0.1705	Min Kurtosis	0.48
Current RMS	0.2095	Current Kurtosis	0.56

Dataset Information

Total Measurements	Initial RUL	Time to Failure
200	55.9 hours	2.3 days

Model Details:

- Algorithm: LightGBM v2 (Optuna-tuned, 80 trials)
- Critical Zone MAE: 2.88 hours (0-50h RUL range)
- Overall MAE: 13.42 hours
- R²: 0.9852 (98.5% variance explained)
- Dataset: NASA IMS Bearing Dataset
- Features: 50 selected from 380 engineered features
- Author: Anudeep | Mechanical Engineer → Data Scientist